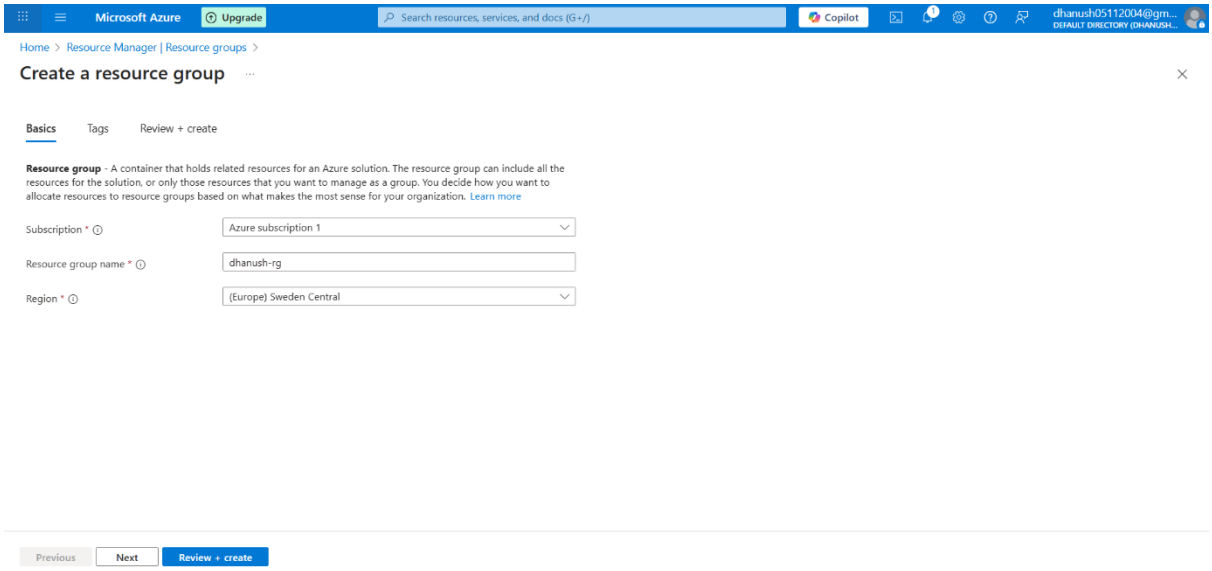
LAB 12 : PRIVATE DNS ZONES

Goal

- **Create a Private DNS Zone in Azure**
- **Link it to a Virtual Network**
- **Add DNS records**
- **Test resolution between two VMs**

Step 1: Create Resource Group

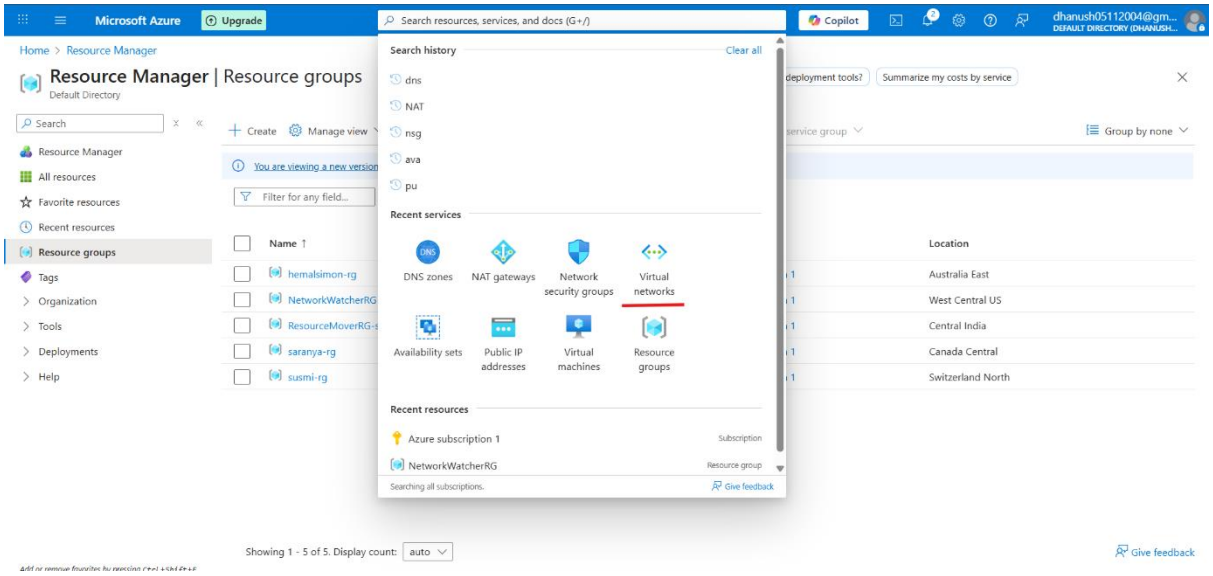
- **Name:** <studentname>_rg
- Example: Arun_rg

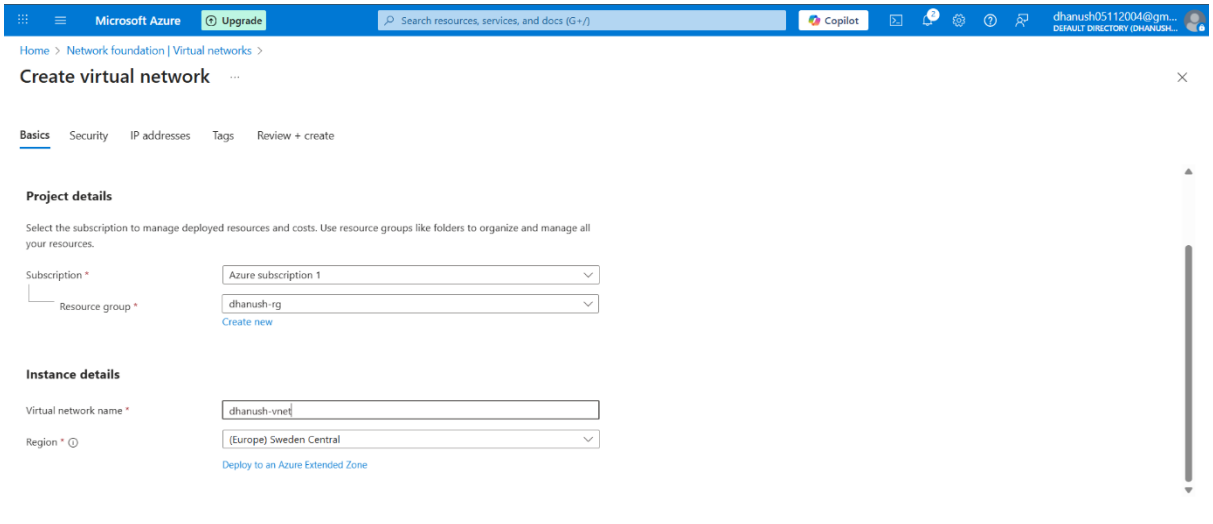


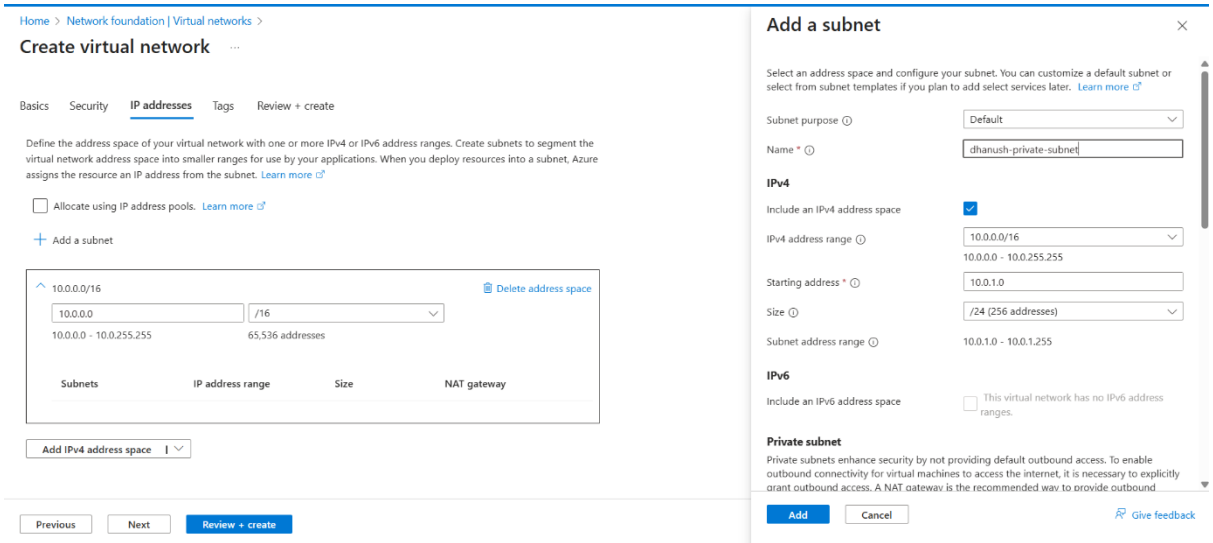
Step 2: Create Virtual Network & Subnet

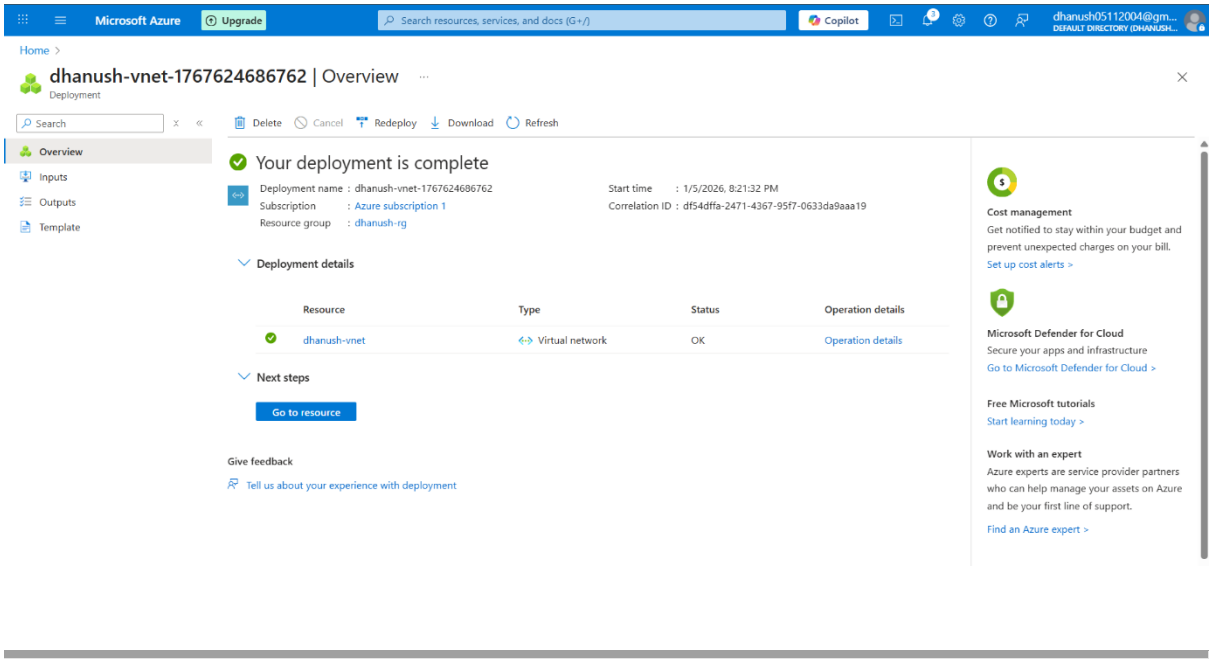
- **VNet Name:** <studentname>_vnet

- Address space: 10.10.0.0/16
- **Subnet Name:** <studentname>_subnet1
- Subnet range: 10.10.1.0/24


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Resource Manager
Resource Manager | Resource groups
Default Directory
Search
Resource Manager
All resources
Favorite resources
Recent resources
Resource groups
Tags
Organization
Tools
Deployments
Help
Create
Manage view
Group by none
Filter for any field...
Name
Location
hemalsimon-rg
NetworkWatcherRG
ResourceMoverRG-s
saranya-rg
susmi-rg
Australia East
West Central US
Central India
Canada Central
Switzerland North
Search history
Clear all
dns
NAT
nsg
ava
pu
Recent services
DNS zones
NAT gateways
Network security groups
Virtual networks
Availability sets
Public IP addresses
Virtual machines
Resource groups
Recent resources
Azure subscription 1
Subscription
NetworkWatcherRG
Resource group
Searching all subscriptions.
Give feedback
deployment tools?
Summarize my costs by service
Showing 1 - 5 of 5. Display count: auto
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Network foundation | Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual network name *
dhanush-vnet
Region *
(Europe) Sweden Central
Deploy to an Azure Extended Zone
Previous
Next
Review + create
Give feedback


Home > Network foundation | Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. Learn more
Allocate using IP address pools. Learn more
Add a subnet
10.0.0.0/16
Delete address space
10.0.0.0
/16
10.0.0.0 - 10.0.255.255
65,536 addresses
Subnets
IP address range
Size
NAT gateway
Add IPv4 address space
Previous
Next
Review + create
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more
Subnet purpose
Default
Name *
dhanush-private-subnet
IPv4
Include an IPv4 address space
IPv4 address range
10.0.0.0/16
10.0.0.0 - 10.0.255.255
Starting address *
10.0.1.0
Size
/24 (256 addresses)
Subnet address range
10.0.1.0 - 10.0.1.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Private subnet
Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound
Add
Cancel
Give feedback

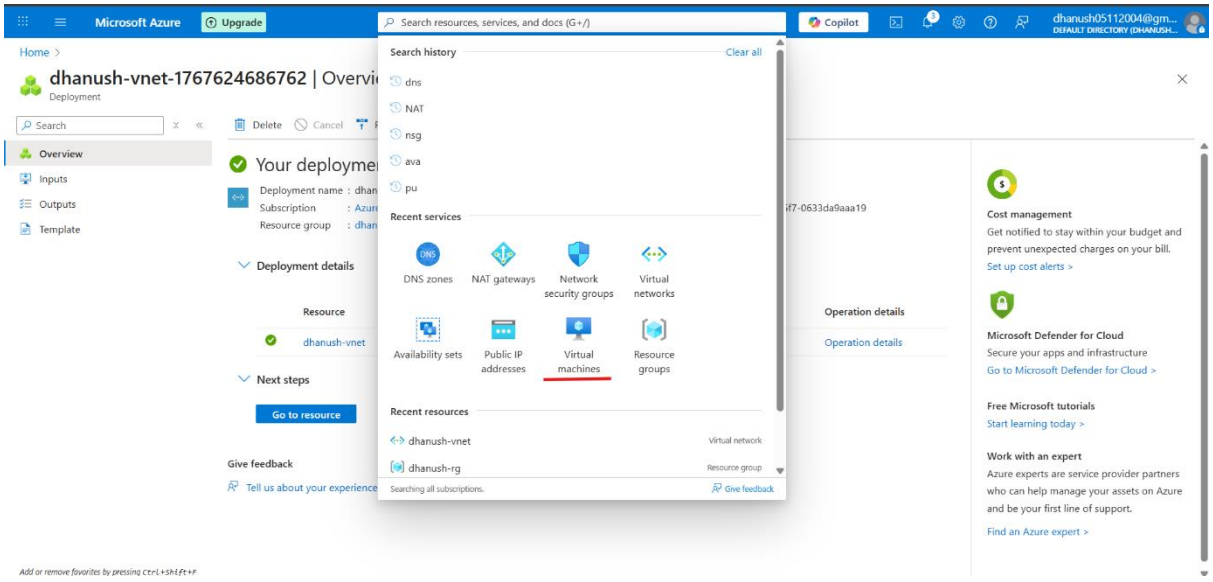


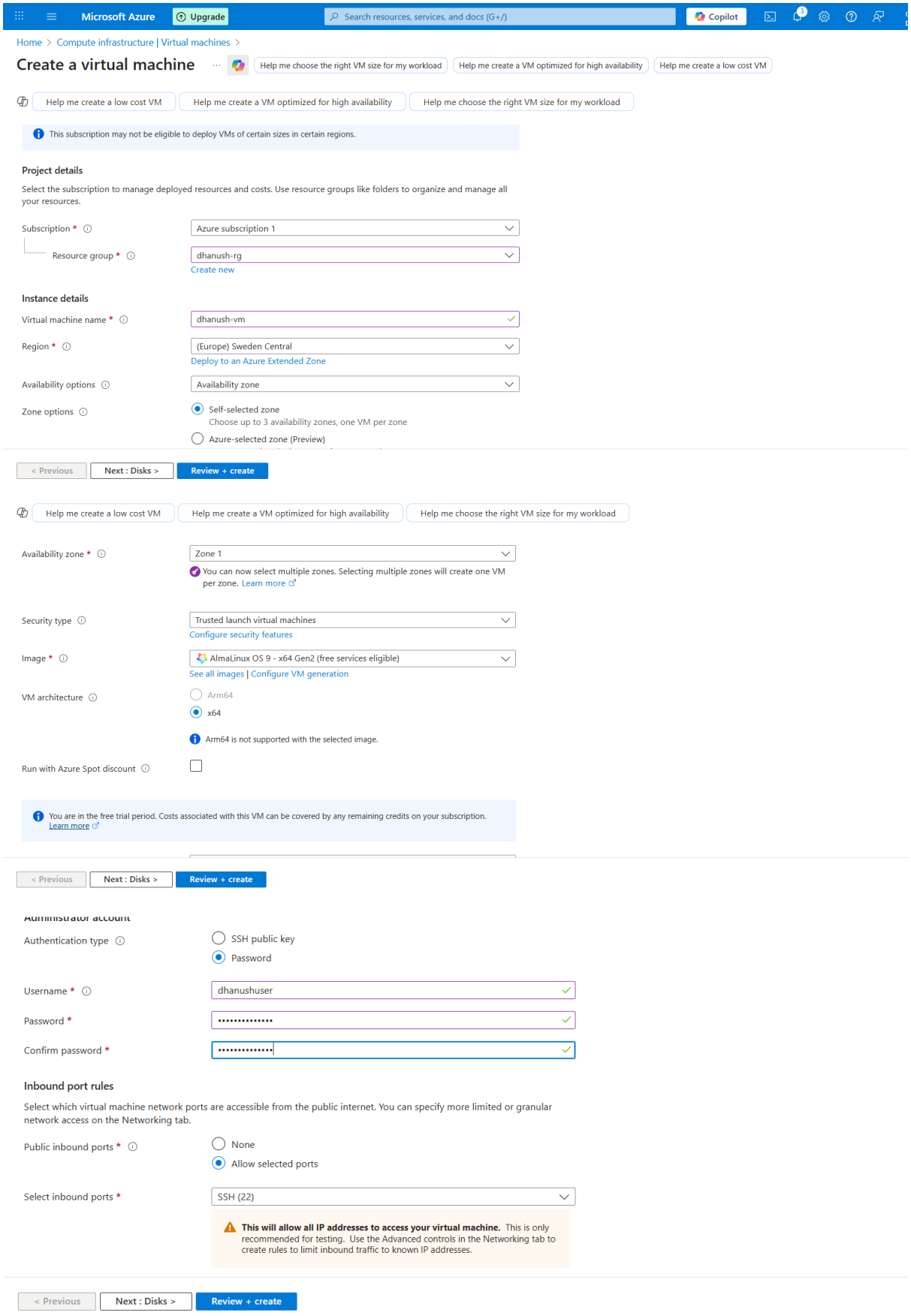
Step 3: Deploy Two VMs

- **VM1 Name:** <studentname>_vm1
- **VM2 Name:** <studentname>_vm2
- Place both inside: <studentname>_vnet → <studentname>_subnet1

✅ Verify connectivity by pinging private IPs between VMs.

VM 1


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home
dhanush-vnet-1767624686762 | Overvi
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Your deployme
Deployment name : dhan
Subscription : Azur
Resource group : dhan
Deployment details
Resource
dhanush-vnet
Next steps
Go to resource
Give feedback
Tell us about your experience
Add or remove favorites by pressing Ctrl+Shift+F
Search history
Clear all
dns
NAT
nsg
ava
pu
Recent services
DNS zones
NAT gateways
Network security groups
Virtual networks
Availability sets
Public IP addresses
Virtual machines
Resource groups
Recent resources
dhanush-vnet
Virtual network
dhanush-rg
Resource group
Searching all subscriptions.
Give feedback
Operation details
Operation details
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a VM optimized for high availability
Help me create a low cost VM
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
This subscription may not be eligible to deploy VMs of certain sizes in certain regions.
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual machine name *
dhanush-vm
Region *
(Europe) Sweden Central
Deploy to an Azure Extended Zone
Availability options
Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
< Previous
Next : Disks >
Review + create
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Availability zone *
Zone 1
You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more
Security type
Trusted launch virtual machines
Configure security features
Image *
AlmaLinux OS 9 - x64 Gen2 (free services eligible)
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription.
Learn more
< Previous
Next : Disks >
Review + create
Administrator account
Authentication type
SSH public key
Password
Username *
dhanushuser
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
Allow selected ports
Select inbound ports *
SSH (22)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
< Previous
Next : Disks >
Review + create

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | **Networking** | Management | Monitoring | Advanced | Tags | Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * : dhanush-vnet
Create new

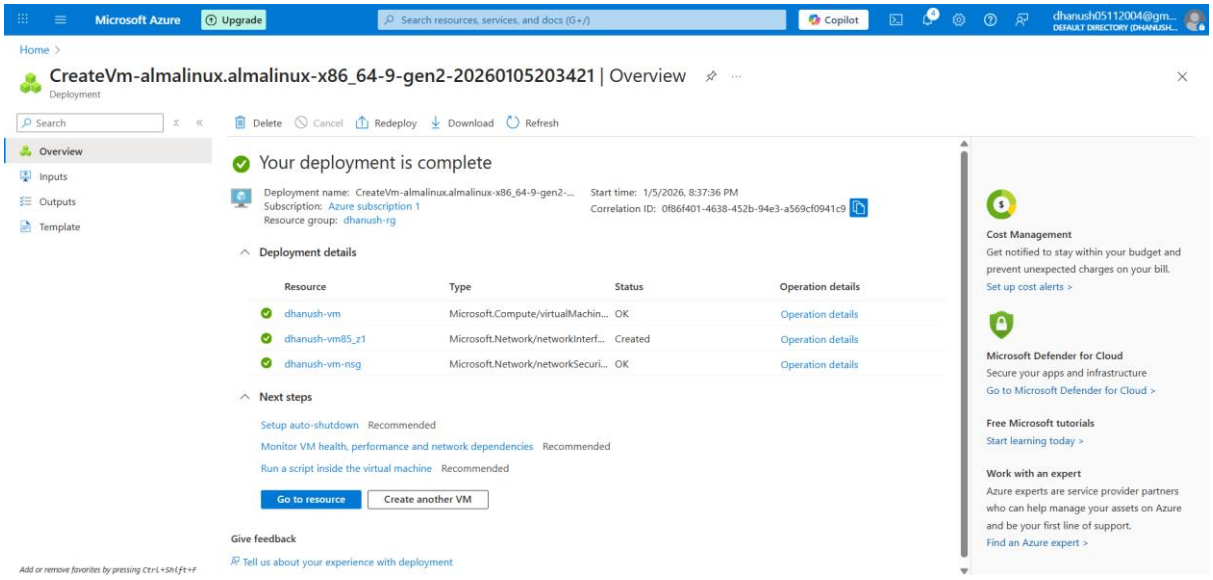
Subnet * : dhanush-private-subnet (10.0.1.0/24)
Manage subnet configuration

Public IP : None
Create new

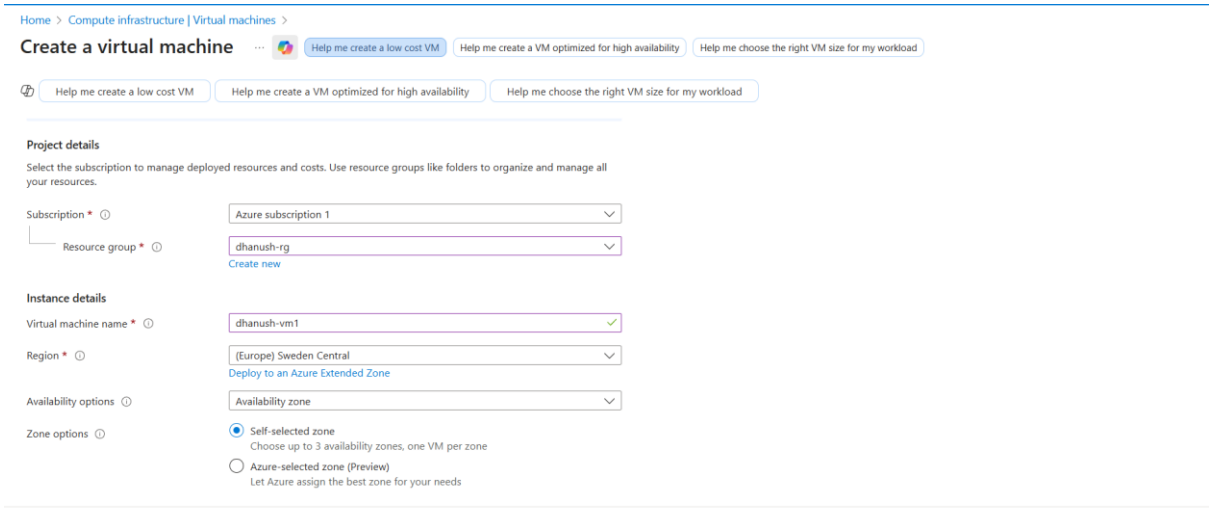
NIC network security group : None / Basic / Advanced

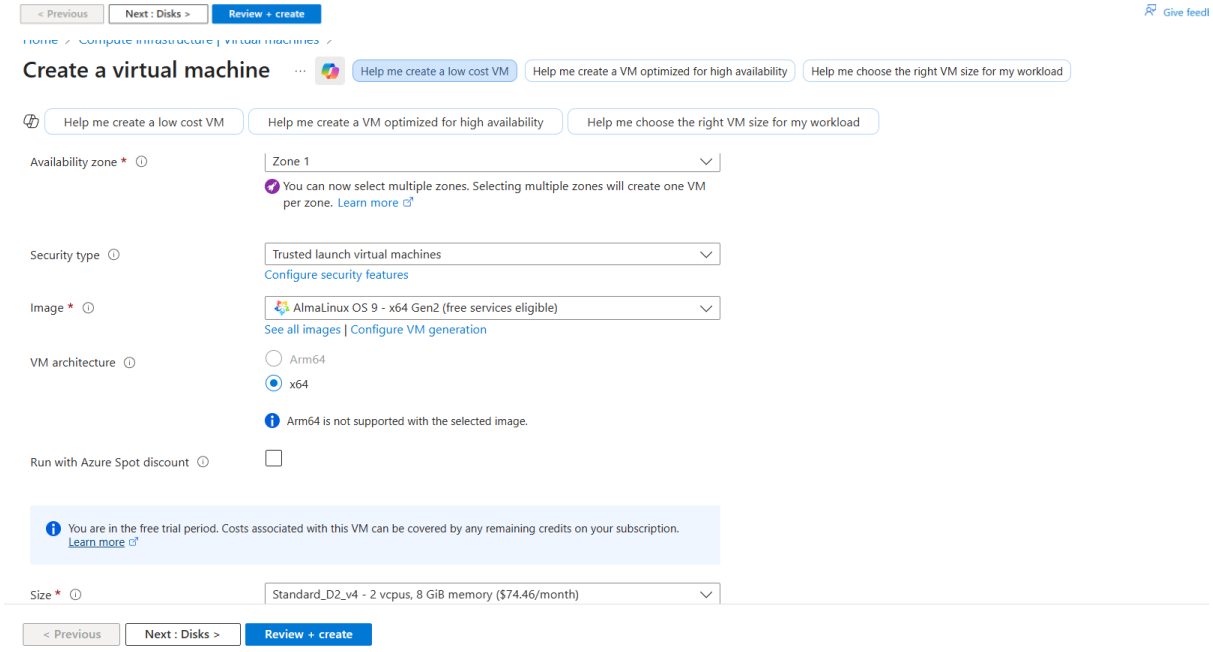
Public inbound ports * : None

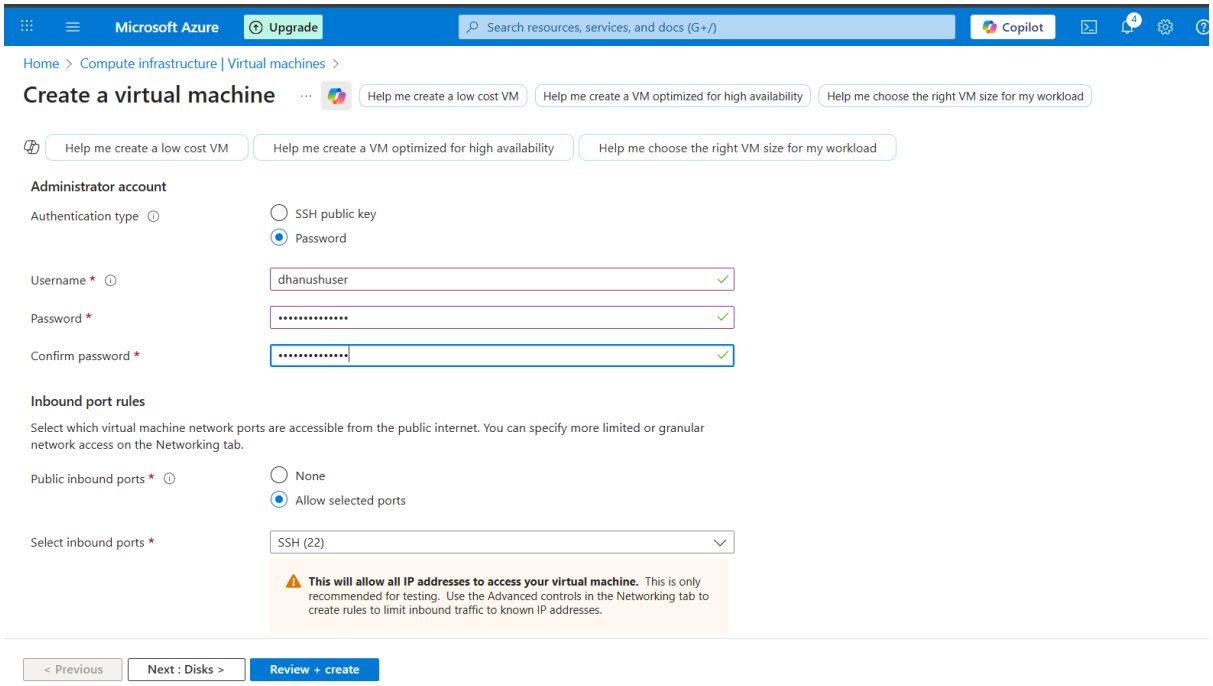
< Previous | Next : Management > | Review + create

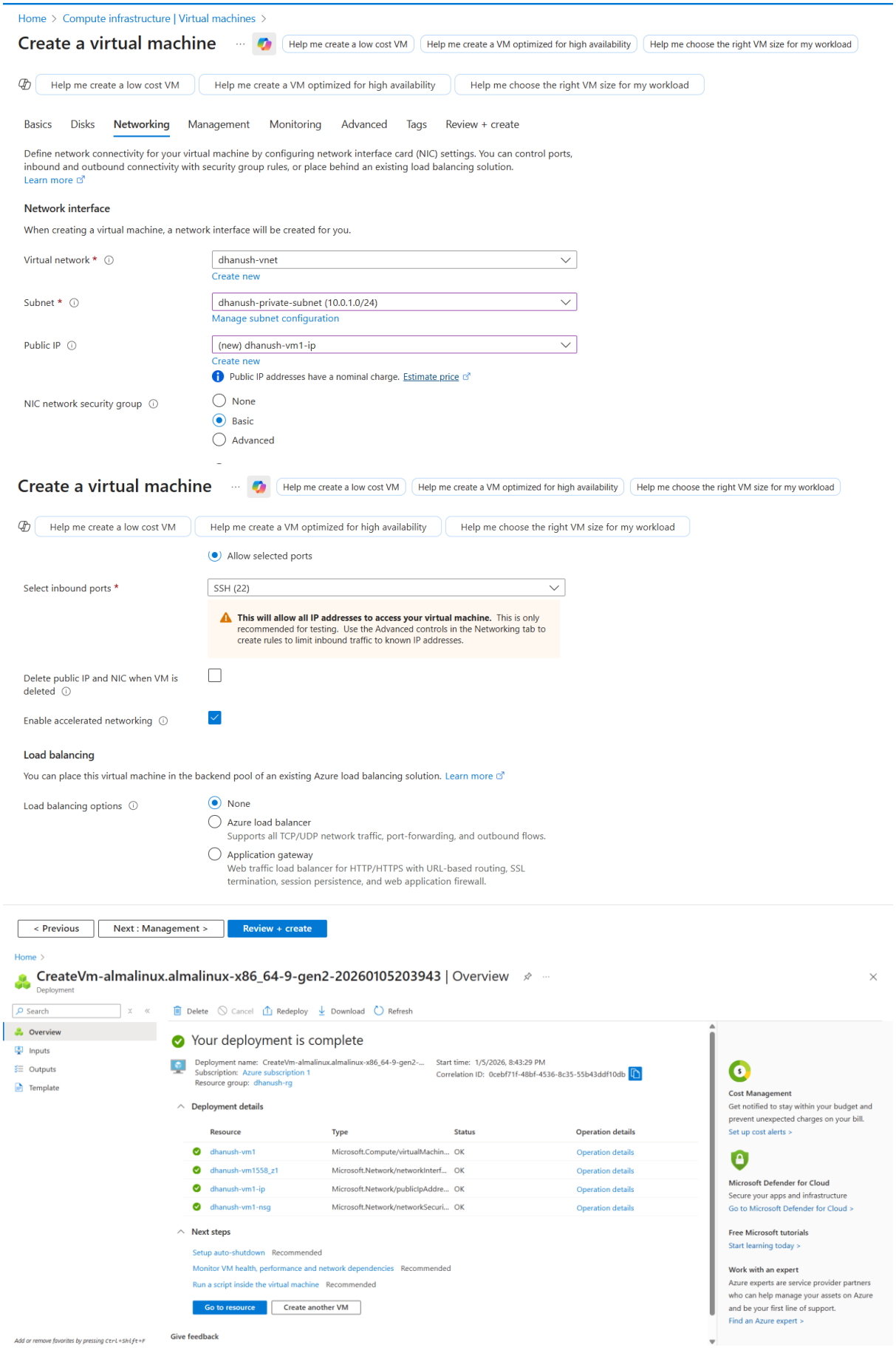


VM 2


Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual machine name *
dhanush-vm1
Region *
(Europe) Sweden Central
Deploy to an Azure Extended Zone
Availability options
Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
< Previous
Next : Disks >
Review + create
Give feedl

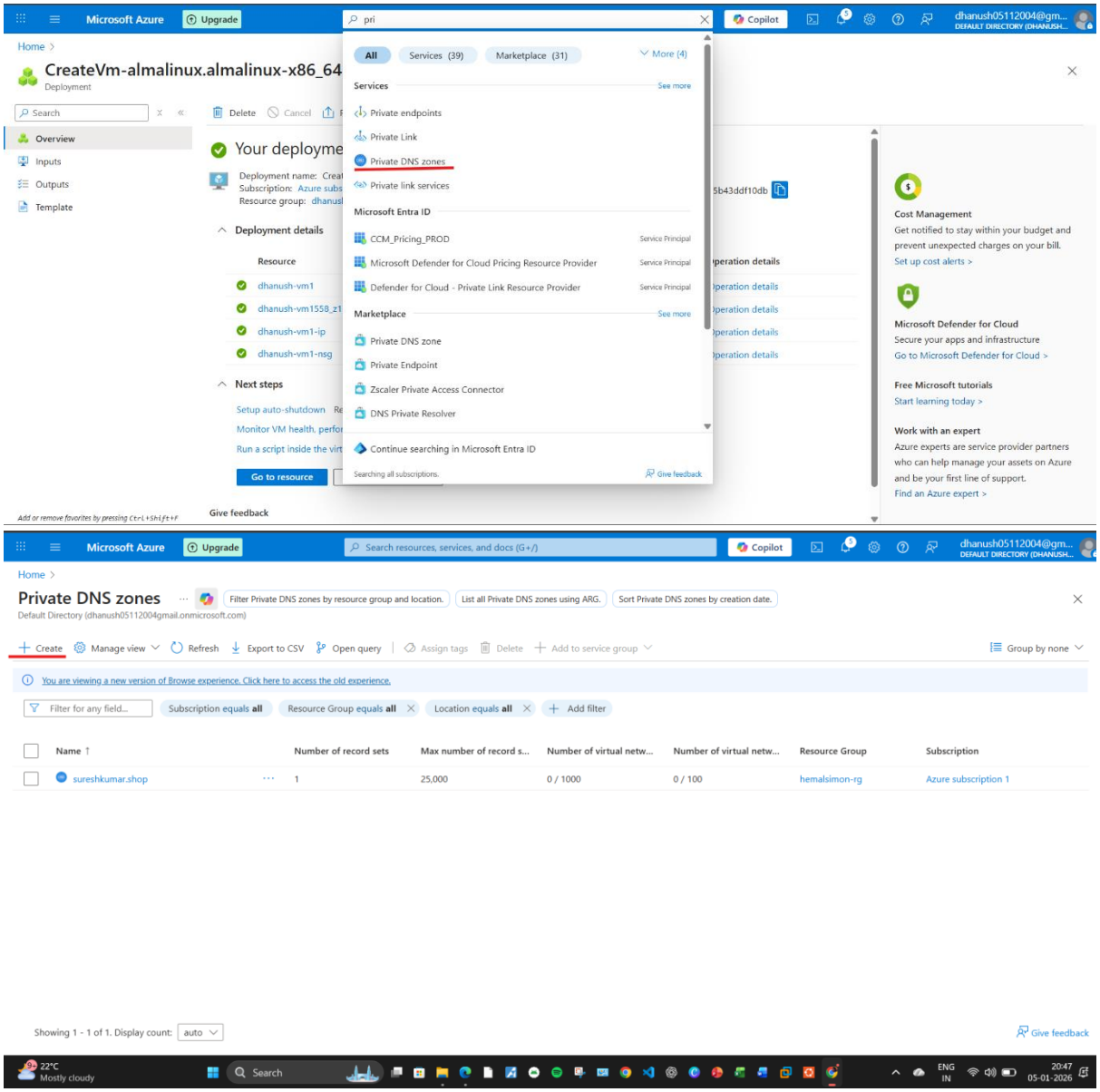

Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Availability zone *
Zone 1
You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more
Security type
Trusted launch virtual machines
Configure security features
Image *
AlmaLinux OS 9 - x64 Gen2 (free services eligible)
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription.
Learn more
Size *
Standard_D2_v4 - 2 vcpus, 8 GiB memory ($74.46/month)
< Previous
Next : Disks >
Review + create

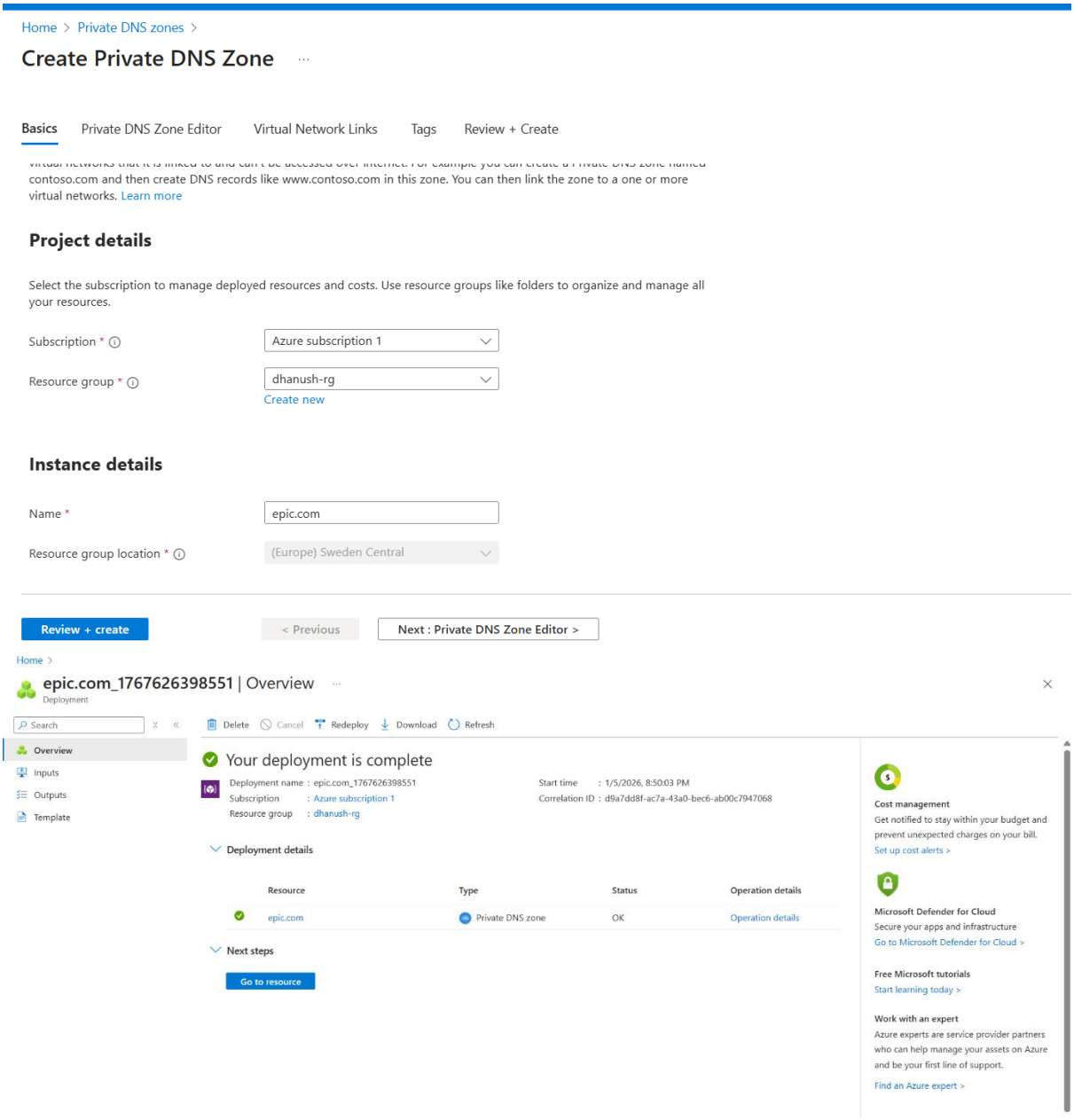

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Administrator account
Authentication type
SSH public key
Password
Username *
dhanushuser
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
Allow selected ports
Select inbound ports *
SSH (22)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
< Previous
Next : Disks >
Review + create


Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Basics Disks Networking Management Monitoring Advanced Tags Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more
Network interface
When creating a virtual machine, a network interface will be created for you.
Virtual network *
dhanush-vnet
Create new
Subnet *
dhanush-private-subnet (10.0.1.0/24)
Manage subnet configuration
Public IP
(new) dhanush-vm1-ip
Create new
Public IP addresses have a nominal charge. Estimate price
NIC network security group
None
Basic
Advanced
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Allow selected ports
Select inbound ports *
SSH (22)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
Delete public IP and NIC when VM is deleted
Enable accelerated networking
Load balancing
You can place this virtual machine in the backend pool of an existing Azure load balancing solution. Learn more
Load balancing options
None
Azure load balancer
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
Application gateway
Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.
< Previous
Next : Management >
Review + create
Home >
CreateVm-almalinux.almalinux-x86_64-9-gen2-20260105203943 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name: CreateVm-almalinux.almalinux-x86_64-9-gen2-...
Subscription: Azure subscription 1
Resource group: dhanush-rg
Start time: 1/5/2026, 8:43:29 PM
Correlation ID: 0cebf71f-48bf-4536-8c35-55b43ddf10db
Deployment details
Resource
Type
Status
Operation details
dhanush-vm1
Microsoft.Compute/virtualMachin...
OK
Operation details
dhanush-vm1558_z1
Microsoft.Network/networkInterf...
OK
Operation details
dhanush-vm1-ip
Microsoft.Network/publicIpAddre...
OK
Operation details
dhanush-vm1-nsg
Microsoft.Network/networkSecuri...
OK
Operation details
Next steps
Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended
Go to resource
Create another VM
Give feedback
Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >
Add or remove favorites by pressing Ctrl+Shift+F

Step 4: Create Private DNS Zone

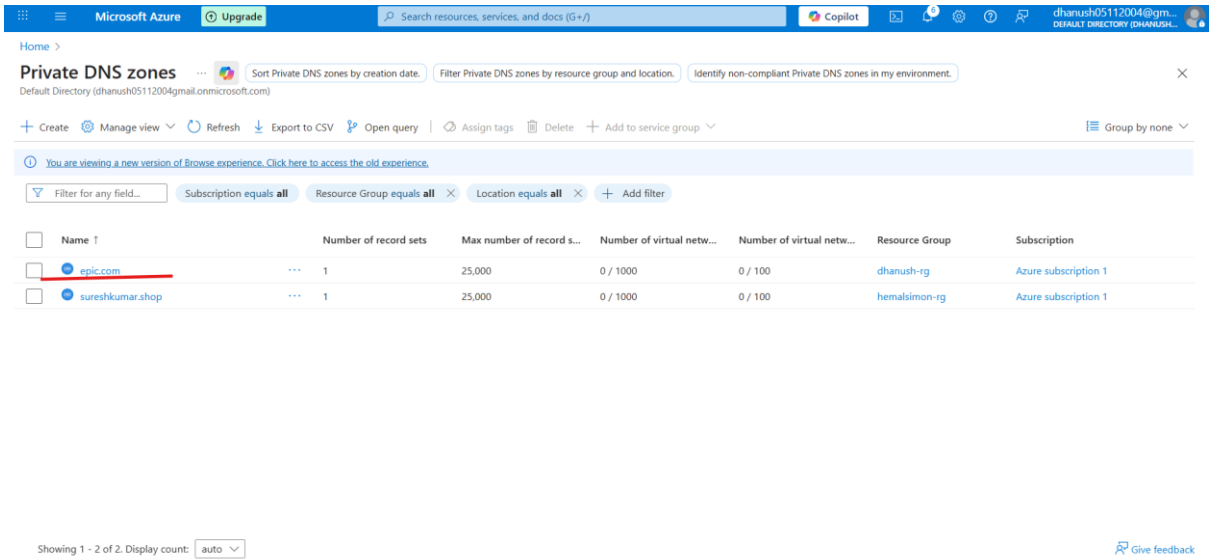
- **Private DNS Zone**
  Name: <studentname>_privatedns.local
- Example: Arun_privatedns.local

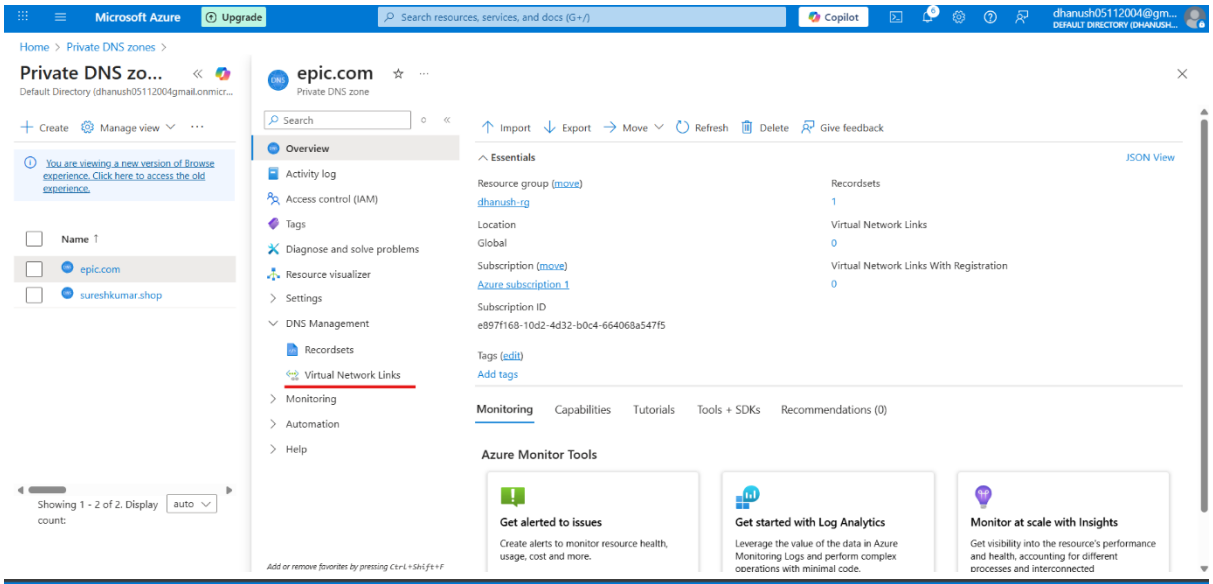


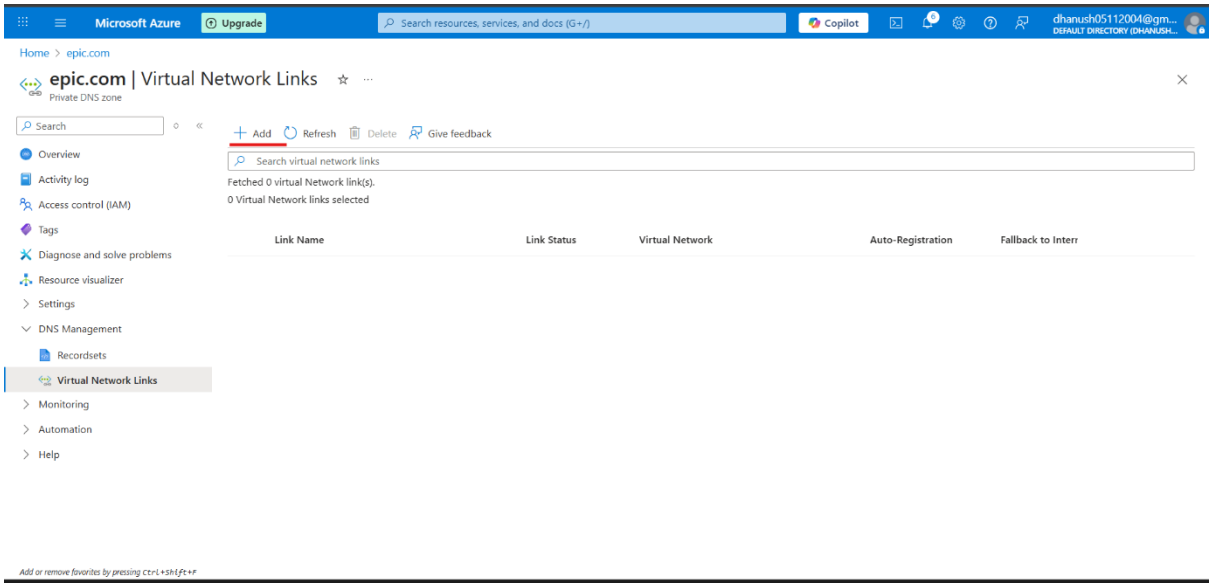


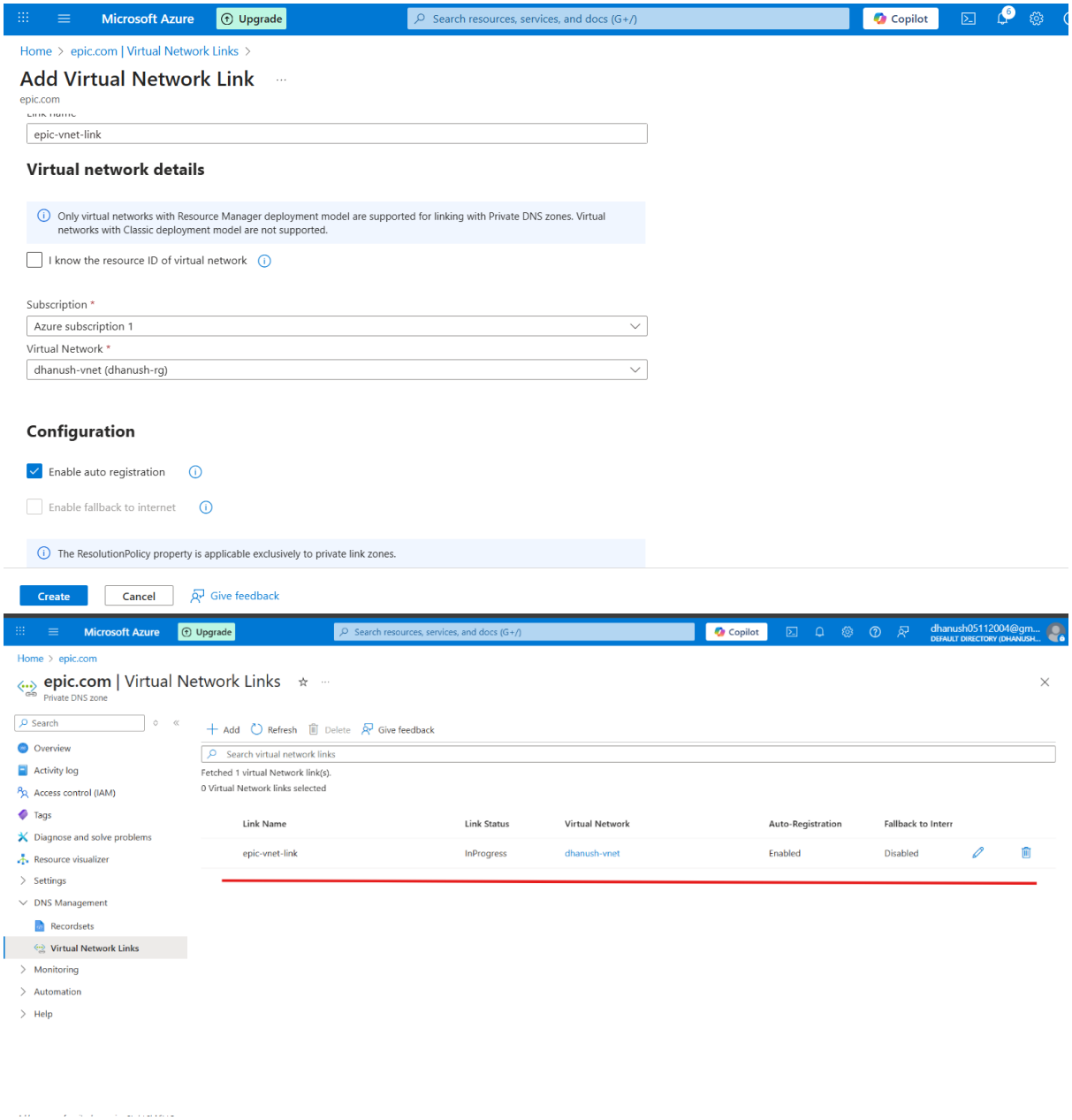
Step 5: Link DNS Zone to VNet

- Inside the DNS Zone → **Virtual Network Links → + Add**
- **Link Name:** <studentname>_vnetlink
- Select <studentname>_vnet
- Enable **Auto-registration** ☑


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home >
Private DNS zones
Default Directory (dhanush05112004gmail.onmicrosoft.com)
Sort Private DNS zones by creation date.
Filter Private DNS zones by resource group and location.
Identify non-compliant Private DNS zones in my environment.
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Delete
Add to service group
Group by none
You are viewing a new version of Browse experience. Click here to access the old experience.
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Number of record sets
Max number of record s...
Number of virtual netw...
Number of virtual netw...
Resource Group
Subscription
epic.com
1
25,000
0 / 1000
0 / 100
dhanush-rg
Azure subscription 1
sureshkumar.shop
1
25,000
0 / 1000
0 / 100
hemalsimon-rg
Azure subscription 1
Showing 1 - 2 of 2. Display count:
auto
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Private DNS zones >
Private DNS zo...
Default Directory (dhanush05112004gmail.onmicr...
Create
Manage view
You are viewing a new version of Browse experience. Click here to access the old experience.
Name
epic.com
sureshkumar.shop
Showing 1 - 2 of 2. Display count:
auto
epic.com
Private DNS zone
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
DNS Management
Recordsets
Virtual Network Links
Monitoring
Automation
Help
Import
Export
Move
Refresh
Delete
Give feedback
Essentials
JSON View
Resource group (move)
dhanush-rg
Location
Global
Subscription (move)
Azure subscription 1
Subscription ID
e897f168-10d2-4d32-b0c4-664068a547f5
Tags (edit)
Add tags
Recordsets
1
Virtual Network Links
0
Virtual Network Links With Registration
0
Monitoring
Capabilities
Tutorials
Tools + SDKs
Recommendations (0)
Azure Monitor Tools
Get alerted to issues
Create alerts to monitor resource health, usage, cost and more.
Get started with Log Analytics
Leverage the value of the data in Azure Monitoring Logs and perform complex operations with minimal code.
Monitor at scale with Insights
Get visibility into the resource's performance and health, accounting for different processes and interconnected
Add or remove favorites by pressing Ctrl+Shift+F

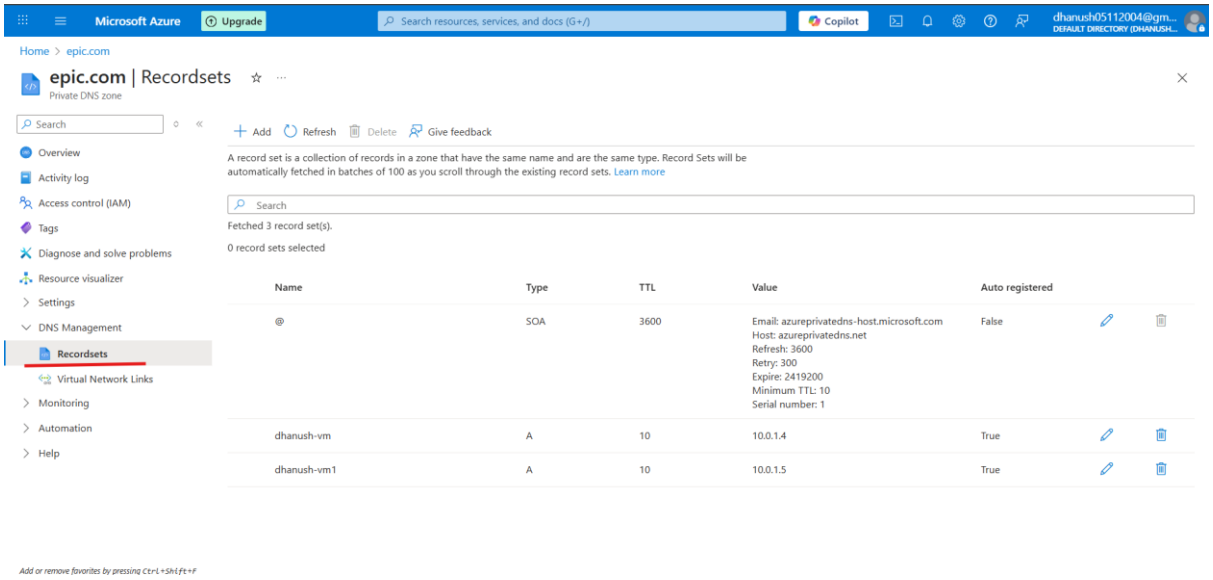

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > epic.com
epic.com | Virtual Network Links
Private DNS zone
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
DNS Management
Recordsets
Virtual Network Links
Monitoring
Automation
Help
Add
Refresh
Delete
Give feedback
Search virtual network links
Fetched 0 virtual Network link(s).
0 Virtual Network links selected
Link Name
Link Status
Virtual Network
Auto-Registration
Fallback to Interr
Add or remove favorites by pressing Ctrl+Shift+F



Step 6: Add DNS Records

1. **A Record**
   - Name: <studentname>_app1
   - IP: Private IP of <studentname>_vm1
2. **CNAME Record**
   - Name: <studentname>_web

- Alias: <studentname>_app1.<studentname>_privatedns.local



Step 7: Test DNS Resolution

From **<studentname>_vm2** run:

nslookup <studentname>_app1.<studentname>_privatedns.local

nslookup <studentname>_web.<studentname>_privatedns.local

ping <studentname>_app1.<studentname>_privatedns.local

☑ Should resolve
to <studentname>_vm1 private IP.

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

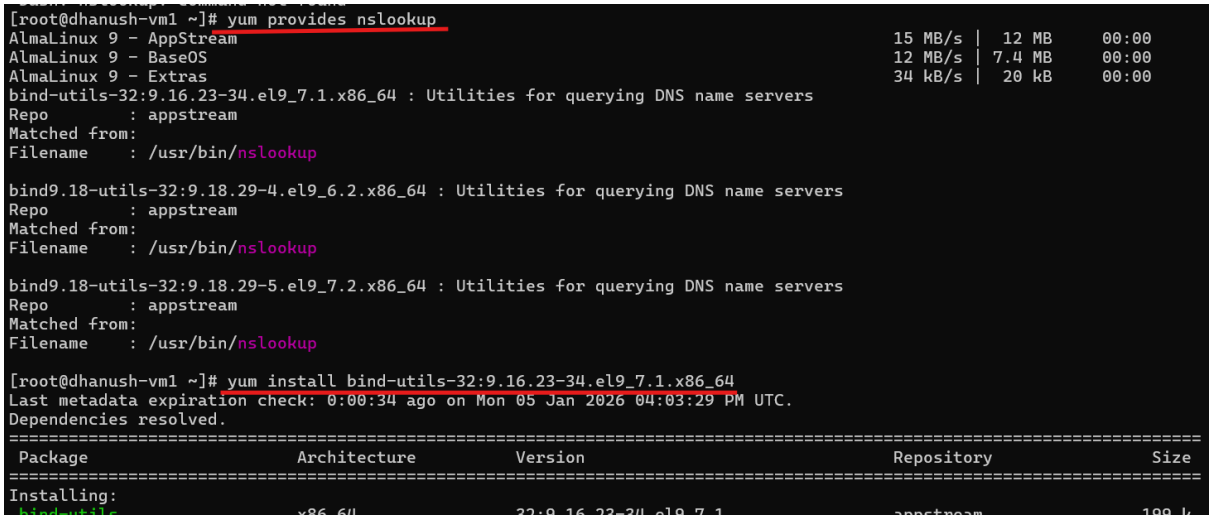
C:\Users\DHANUSH>ssh dhanushuser@9.223.176.61
dhanushuser@9.223.176.61's password:
Last login: Mon Jan  5 15:54:42 2026 from 10.0.1.4
[dhanushuser@dhanush-vm1 ~]$ ssh dhanushuser@dhanush-vm.epic.com
dhanushuser@dhanush-vm.epic.com's password:
Last login: Mon Jan  5 15:54:13 2026 from 10.0.1.5
[dhanushuser@dhanush-vm ~]$ sudo -i
[root@dhanush-vm ~]# ssh dhanushuser@dhanush-vm1.epic.com
dhanushuser@dhanush-vm1.epic.com's password:
Last login: Mon Jan  5 15:58:37 2026 from 203.192.204.163
```

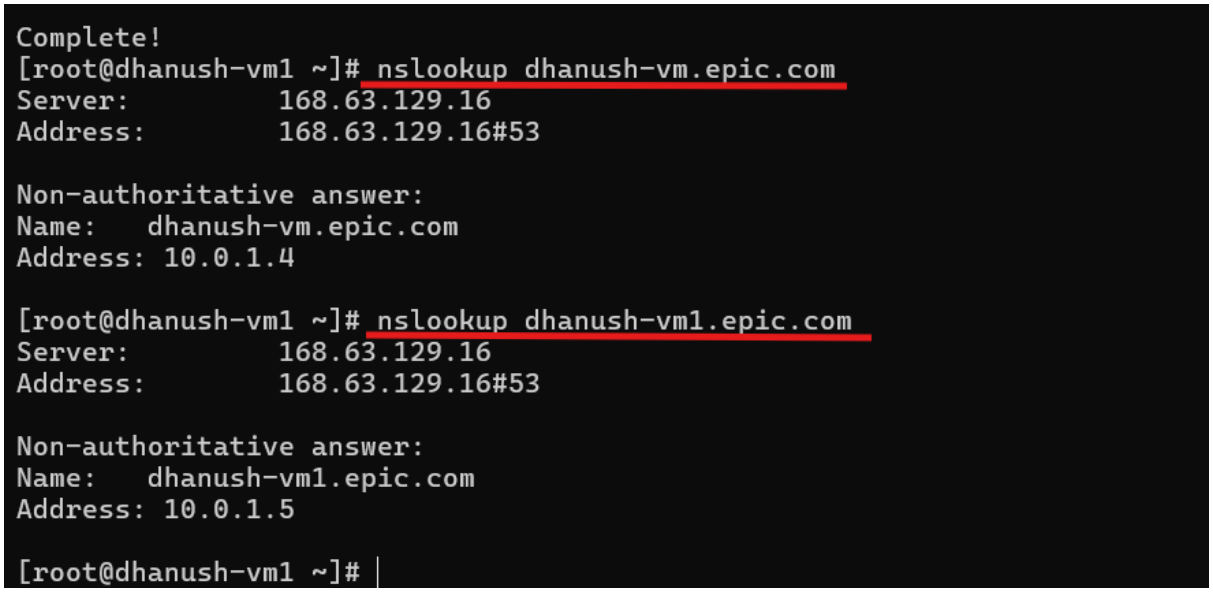
```
[root@dhanush-vm1 ~]# yum provides nslookup
AlmaLinux 9 - AppStream                                          15 MB/s |  12 MB     00:00
AlmaLinux 9 - BaseOS                                             12 MB/s | 7.4 MB     00:00
AlmaLinux 9 - Extras                                             34 kB/s |  20 kB     00:00
bind-utils-32:9.16.23-34.el9_7.1.x86_64 : Utilities for querying DNS name servers
Repo        : appstream
Matched from:
Filename    : /usr/bin/nslookup

bind9.18-utils-32:9.18.29-4.el9_6.2.x86_64 : Utilities for querying DNS name servers
Repo        : appstream
Matched from:
Filename    : /usr/bin/nslookup

bind9.18-utils-32:9.18.29-5.el9_7.2.x86_64 : Utilities for querying DNS name servers
Repo        : appstream
Matched from:
Filename    : /usr/bin/nslookup

[root@dhanush-vm1 ~]# yum install bind-utils-32:9.16.23-34.el9_7.1.x86_64
Last metadata expiration check: 0:00:34 ago on Mon 05 Jan 2026 04:03:29 PM UTC.
Dependencies resolved.
================================================================================
 Package                Architecture          Version                  Repository                Size
================================================================================
Installing:
 bind-utils             x86_64                32:9.16.23-34.el9_7.1    appstream                 199 k
```





Step 8: Verify Auto-Registration

- On <studentname>_vm1, check hostname:

hostname

- Then from <studentname>_vm2:

nslookup <hostname>.<studentname>_privatedns.local

Expected Outcome

- All DNS queries resolve within the VNet using the Private DNS Zone.
- Naming convention makes resources **organized and student-specific**.